

Decision Policy: Should We Fix It?

Individual component of FaithGuard · Sharaf · Revised plan, 8 October 2026

This document describes one component of FaithGuard, a layer that checks AI answers about financial reports against the evidence they were given, and then sends each answer, fixes it, or withholds it with a reason. This component makes the decision: for each answer it predicts what sending, repairing or abstaining would produce, chooses an action, and certifies the error rate among the answers it lets through.

Nothing in this document has been built or measured yet. Every result below is a hypothesis with a stated way to reject it. All example numbers are illustrative, not results.

1 Purpose

This component tests whether **predicting each action's outcome** beats a plain risk threshold under a certified error limit, and what happens to that policy when the repair tool is upgraded.

Working title: Should We Fix It? Certified Accept–Repair–Abstain Policies for Financial QA and Their Fragility When the Repairer Changes.

For every answer, the component:

1. estimates **P(unsafe)**, the chance the original answer is unsafe to return;
2. estimates **P(fix)**, the chance repair returns a safe, useful answer;
3. chooses to send, repair or abstain with thresholds certified so that the error rate among sent answers stays at or below 10%.

It also owns the shared infrastructure that makes the three papers comparable: the record format, the replay harness, the split manifest, the statistics scripts and the answer-generation runs.

Example

The question is “What was the Group’s profit after tax in FY2025?” The AI answers: “Group profit after tax rose 7.0% to Rs. 12,450 million.” Detection flags the number as the Bank’s figure at the wrong scale. The policy sees that the correct Group figure is in the table, so P(fix) is high. It chooses repair. Had the Group figure been missing, P(fix) would be low and the policy would abstain, showing the evidence and a reason.

2 Position in the system

	Content
Receives	One signal record per answer from detection, with pre-action features only. Offline, the stored outcomes of every action on every item.
Produces	One action per answer (send, repair or abstain), a reason code when it abstains, and the certified thresholds (τ_1 , τ_2). Offline, the replay dataset and the transfer results.
Does not do	It does not detect spans (the detection component does) and does not fix text (the repair component does).

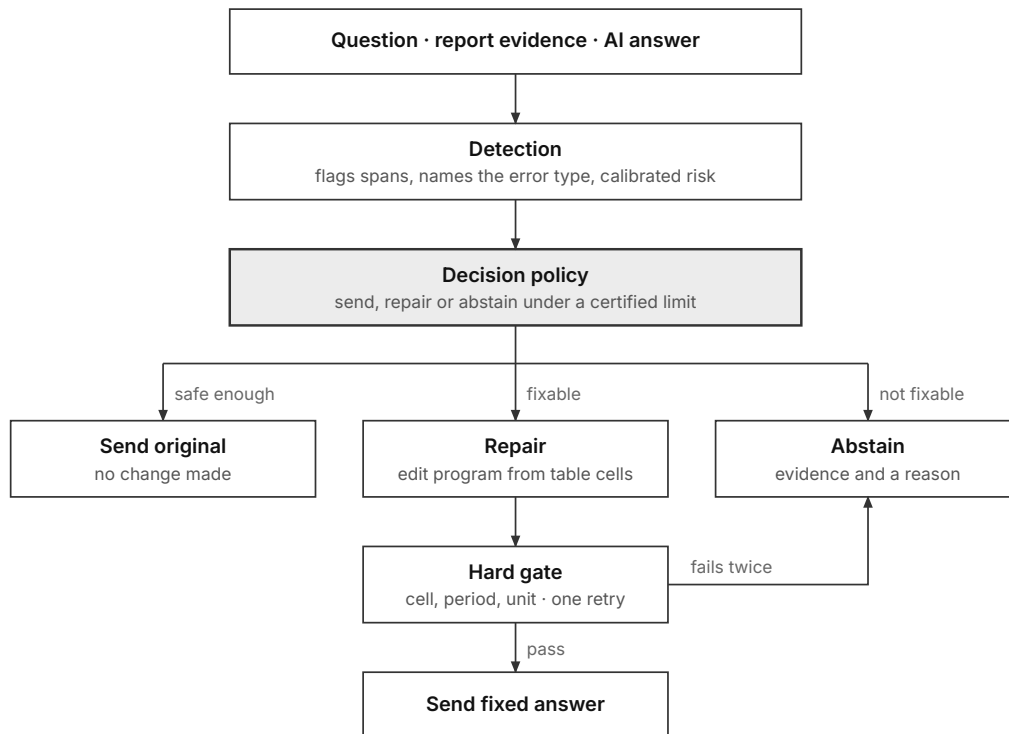


Figure 1. Where the component sits in FaithGuard. The shaded box is this component.

3 Research questions

1. At the same certified risk ($\alpha = 0.10$), does an outcome-aware policy return more correct, useful answers than a calibrated detector threshold or a fixed strategy?
2. Can the error rate among returned answers be certified for the whole pipeline, including repair-introduced errors, at sample sizes a student team can label?
3. When the repairer changes from v0 to v1, does the old policy stay safe? Is recalibration enough, or must it be refitted?

4 Component architecture

Two outcome models feed one decision rule. The replay dataset and certification run offline.

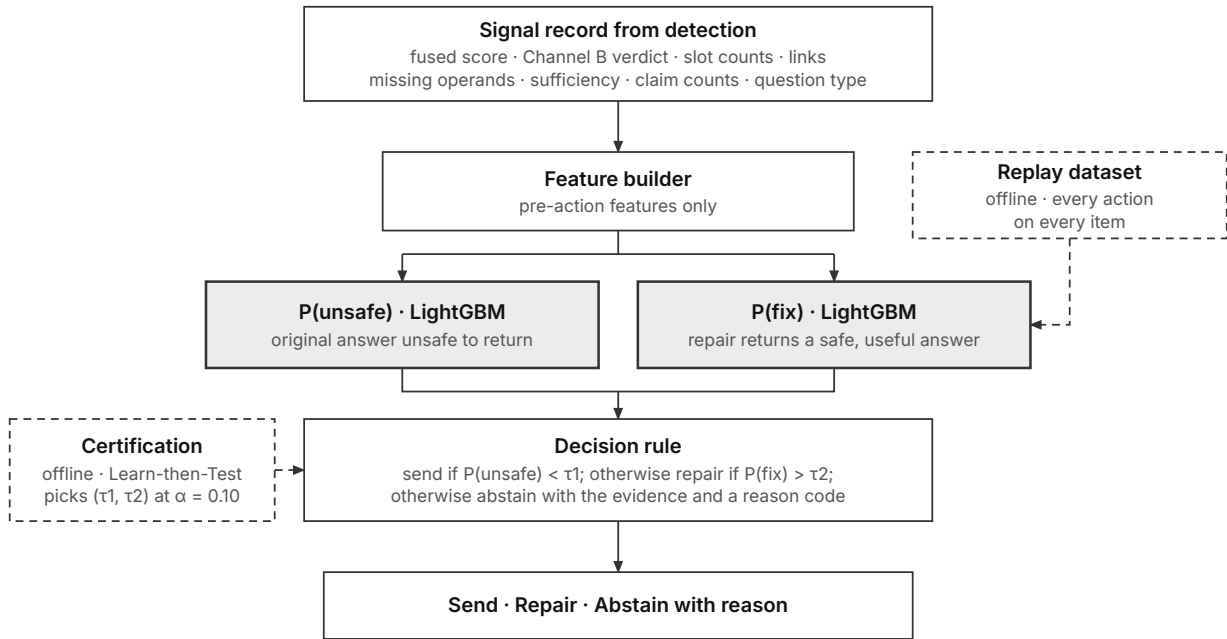


Figure 2. Internal architecture. The two outcome models (shaded) are what this component trains. Dashed boxes run offline.

Replay

Keep, repair and abstain are run on every item, and each outcome is scored. A 3×3 flip matrix records each answer's state before and after repair. Answers that repair made worse are counted, not hidden inside a net gain.

State (before and after repair)	Meaning
Unsupported	The evidence does not support the answer
Supported but unhelpful	Supported by the evidence, but does not usefully answer the question
Supported and useful	Supported by the evidence and answers the question

Features

Pre-action only: fused detector score, Channel B verdict, failing-slot counts, link count, missing operands, an evidence-sufficiency check, claim counts and question type.

Two outcome models (LightGBM)

- **P(unsafe)**: the chance the original answer is unsafe to return;
- **P(fix)**: the chance repair returns a safe, useful answer.

Decision rule

Send the original if $P(\text{unsafe}) < \tau_1$; otherwise repair if $P(\text{fix}) > \tau_2$; otherwise abstain with the evidence and a reason code. A small grid of (τ_1, τ_2) is pre-registered.

Certification

Learn-then-Test over the grid, with fixed-sequence testing and Hoeffding–Bentkus p-values, certifies the linearised loss below on an issuer-disjoint calibration set. SCoRE (the score-select package) is the 2026 comparator.

$$\mathbb{E} [\mathbf{1}\{\text{unsafe emitted}\} - \alpha \cdot \mathbf{1}\{\text{substantive emitted}\}] \leq 0$$

This is the ratio “wrong answers ÷ returned answers $\leq \alpha$ ”, rewritten as a per-item loss that can be certified.

Certification-budget curve

How many calibration items are needed to certify $\alpha = 0.10$ at a given coverage. With zero errors among n independent returned answers, the 95% upper bound is $1 - 0.05^{1/n}$, so about 29 answers are needed to get below the primary α of 10%, and about 59 to get below 5%. This is a practical finding for any team with a small label budget.

Challenger model

TabICLv2, a tabular foundation model that runs on CPU, against LightGBM.

5 Data and shared foundation

All three components use one benchmark, built once and frozen, so their results can be compared and combined.

Source	Role	Labels	Licence
RAGTruth	General span training for detection	Human spans	MIT
FinQA	Gold programs for repair targets; injected errors	Gold answers and programs	MIT
TAT-QA	Table-and-text arithmetic targets	Gold answers	CC BY 4.0
US XBRL facts (SEC EDGAR)	Automatic wrong-context negatives; operand checks	Automatic	Public filings
Team benchmark, US and Sri Lanka	Main natural-error test, calibration set, policy outcomes	Human, blinded	Team release terms

FinRank (fresh 2024–25 US questions, CC BY-NC 4.0) is the one optional external test.

Answers and evidence

- **Two generators:** Qwen3.5-9B and Gemma 4 12B-it, both Apache-2.0, run once at 4-bit with llama.cpp on a free T4 and stored. Two vendors are enough for a leave-one-generator-out test.

- **Questions:** 150 US and 250 Sri Lankan question groups from about 12 held-out issuers per country. Each question is written with its gold answer, gold table cells and question type, so code can check numbers later.
- **Evidence:** the main test uses hand-corrected tables, so table-reading errors cannot be mistaken for model errors. One automatic-extraction run, using the best of three tools on 20 tables, is reported as the real-world condition.
- **Retrieval is frozen.** Every method sees the same evidence for the same question.

Rules that keep results honest

- Training, development, calibration and test splits are issuer-disjoint: one company never appears in two splits.
- The test set is locked before any tuning. Hypotheses, the risk target ($\alpha = 0.10$), splits and primary comparisons are pre-registered before the test set is opened.
- One versioned record format is shared by all components. Gold labels live in a separate store that no component reads at decision time.
- Intervals use a bootstrap clustered by issuer. Counts of issuers, questions and answers are always reported separately.
- LLM judges are used for silver training labels only, never for test labels.

Shared tasks owned by this member

- The record format, replay harness, split manifest and statistics scripts.
- The answer-generation runs, on this member's GPU quota, since the policy itself needs no GPU.
- With the other two members: Sri Lankan report collection and table correction (split by sector), and writing about 133 question groups with gold answers.

6 Experiments

Comparison	Question it answers
Calibrated detector threshold (send or abstain)	RQ1: the simplest certified policy
Always keep; always repair	Fixed strategies
Static four-state reparability classifier	Outcome prediction vs static labels
Stored-outcome oracle	Upper bound and regret
LightGBM vs TabICLv2	Does a foundation model help on a few hundred items?
v0 policy on v1 outputs as-is, recalibrated, refitted	RQ3: the transfer study
Train on Qwen answers, test on Gemma answers	A second, generator shift

Metrics: emission coverage, residual risk, useful coverage, the flip matrix, regret against the oracle, Brier score and ECE per model, and residual risk by group (numeric vs narrative, US vs Sri Lanka).

7 Hypotheses

Hypothesis	Confirmed if	Rejected if
H1. The outcome-aware policy beats the threshold policy	Higher useful coverage at equal certified α	No gain; the simpler policy is recommended
H2. $\alpha = 0.10$ is certifiable at useful coverage	Certificate holds with coverage above the threshold policy	Not reachable; the budget curve says how many labels would be needed
H3. The v0 policy is unsafe on v1; recalibration restores safety	Risk above α as-is, within α after recalibration	Safe as-is, or refitting is needed; both are findings

8 Scope and compute

- **Never waits for teammates.** From November, development runs on a controlled track: FinQA and TAT-QA items with injected errors, where the executor scores outcomes automatically. Detector v0 is released LettuceDetect plus Channel B rules. Repairer v0 is rule-only. Repairer v1 is the trained model, or a prompt-only model if training is late.
- **Minimum publishable result:** the replay dataset, outcome-aware vs threshold policy under certified $\alpha = 0.10$, and the v0-to-v1 transfer study (as-is vs recalibrated) on the human-labelled natural test.
- **Stretch:** Conformal Policy Control as a fourth transfer arm; a separate harm head; a compute-aware decision rule.
- **Compute:** everything runs on a laptop CPU. This member's free GPU quota is therefore used for the shared answer generation and as a backup for the other two members.

Work on this member's quota	Rough GPU hours
Answer generation (about 1,000 answers including the pilot) and zero-shot 9B programs	10–15
Share of pilots, debugging and reruns (15–25 hours across the team)	—

These figures are rough guesses, not measurements. Answers are generated once and saved to a Kaggle Dataset or Google Drive as soon as each step finishes.

9 Relation to existing work

Defensible claim: a paired replay dataset in which every action is run on every item, with harm labelled. On it, an outcome-aware policy is compared with static thresholds under a Learn-then-Test certificate that counts errors repair introduces. The paper then studies how the policy transfers when the repairer changes. The 7 October 2026 check found no paper that measures this.

Claims to avoid: the first accept-repair-abstain policy (CORA and Release Control exist), the first risk-controlled abstention, and calling the replay study causal inference.

Realistic first target: FinNLP 2027, ACM ICAIF 2027 or SLIIT ICAC 2027.

10 Risks and cut list

Risk	Early warning	Fallback
$\alpha = 0.10$ is not certifiable	Feasibility bound fails on the calibration size	Report the certification-budget curve as the finding
Generators make very few errors	Pilot error rate below about 10%	Report it as a finding that bounds the benefit; add harder question types
A teammate's part is late	A checkpoint is missed	Released LettuceDetect plus rules as the detector; rule-only or prompt-only repairer
A new paper overlaps the contribution	Monthly literature refresh	Narrow the claim and cite it; never hide it
Free GPU quota is cut or runs out	Hours used pass the budget	Precomputed outputs; switch between Kaggle and Colab; apply the cut list

Cut list items for this component, in team order: TabICLv2 challenger (1st), generator-shift test (6th). **Never cut:** the human-labelled natural test set and the certified risk study.

11 Plan

By	Milestone
18 Oct 2026	Pilots done; GPU quota recorded
31 Oct 2026	Thin end-to-end demo on 20–40 items, rules only
19 Oct – 30 Nov 2026	Replay harness and the controlled track
by 14 Nov 2026	Pilot answers generated (60 questions $\times$ 2 generators)
1 Dec 2026 – 31 Jan 2027	Outcome models and certification
16 Dec 2026 – 31 Jan 2027	Main answer generation
31 Jan 2027	Test set locked; pre-registration filed
16 Feb – 31 Mar 2027	v0-to-v1 transfer study
31 May 2027	Paper draft
30 Jun 2027	Final report and demo

References

1. A. Angelopoulos et al. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. arXiv:2110.01052, 2021.
2. G. Ke et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 2017.
3. Z. Chen et al. FinQA (EMNLP 2021); F. Zhu et al. TAT-QA (ACL 2021).

CORA, Release Control, SCoRE, TabICLv2, Hoeffding–Bentkus bounds and Conformal Policy Control are named in the text; their full citations are added once each source has been opened and logged in the team’s related-work notes.